

ATHARVA WAGH

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SUMMARY

Software Engineering graduate student with 2+ years of experience building scalable backend, full-stack, and data-intensive systems used by real users. Strong background in Python, C#, TypeScript, SQL, distributed systems, and data analysis, with hands-on experience in analytics pipelines, experimentation, and cloud-native architectures. Comfortable working in agile, cross-functional teams and passionate about using data and engineering to improve user experiences at global scale.

TECHNICAL SKILLS

Languages: Python, C++, C#, Java, TypeScript, JavaScript, SQL, Bash

Development Frameworks: .NET, Next.js, VueJS, React, Angular, Node.js, Django, Springboot

ML/Data Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, PySpark, SciPy, NumPy, Pandas, Matplotlib

Databases: PostgreSQL, MongoDB, MySQL, Firebase, Cassandra, AWS Redshift, SQL Server

Devops & Cloud: Git, GitHub, CI/CD Pipelines, Docker, Kubernetes, Linux, Terraform, AWS (EC2, S3, IAM, SageMaker), Azure, GCP, Firebase, Digital Ocean

Machine Learning & AI: Statistical Modeling, Machine Learning, Deep Learning, Model Optimization, LLMs, RAG Pipelines, Multi-Agent Systems, Context Engineering, Information Retrieval, Data Analytics

System Design: Distributed Systems, Data Pipelines, RESTful APIs, Schema Design, Scalable Architectures

EDUCATION

– **University of Southern California, Los Angeles, CA**

Master of Science in Computer Science

GPA: 3.3/4.0

Aug 2025 - May 2027

– **University of Mumbai, Mumbai, India**

Bachelor of Technology in Information Technology (Blockchain Honors)

GPA: 4.0/4.0

Aug 2020 - May 2024

PROFESSIONAL EXPERIENCE

– **Software Engineer / Consultant [Part Time]**

ASPR Technologies

Leicester, England

Jul 2025 - Present

- * Developed AI-powered automation systems using AWS SageMaker and .NET to enhance CRM communication.
- * Designed and developed scalable, distributed backend services using Python, C++, .NET, Docker, and Kubernetes, supporting enterprise-grade automation workflows.
- * Participated in iterative development, A/B testing, and debugging to continuously improve engineering practices.

– **Software Developer [Full Time]**

Institute of Management and Foreign Studies (IMFS)

Mumbai, India

Jul 2024 - May 2025

- * Developed and maintained backend services and internal automation tools supporting enterprise CRM workflows using Python and .NET APIs
- * Built and debugged containerized backend services using Docker and Kubernetes, improving system reliability and reducing response times by 83%.
- * Diagnosed and resolved production bugs by analyzing logs, tracing existing codebases, and validating fixes in staging environments.

– **Web Developer Intern [Co-Op]**

Mabella SkinCare

Mumbai, India

Jun 2023 - May 2024

- * Engineered backend APIs and internal dashboards for a CRM platform using .NET, PostgreSQL, and AWS
- * Built and optimized database schemas and queries, improving analytics query performance and operational visibility.
- * Implemented systematic testing, debugging, and performance profiling to improve application reliability.

ACADEMIC PROJECTS

– **LLM Powered Interview Officer**

LLM Agents | LangChain | Web Scraping | RAG | Docker | WebSockets | React

2024

- * Developed backend services using Python and FastAPI to support real-time, WebSocket-based interactions
- * Deployed containerized services with Docker Compose and maintained internal APIs for multi-agent workflows. Leveraged AI-assisted development tools to explore unfamiliar codebases and accelerate feature delivery.

– **Blockchain-Based Renewable Energy Trading Platform (Research)**

C++ | Distributed Systems | Capstone Project | Real-time Systems | Embedded Systems

2023-24

- * Built backend control and orchestration services integrating real-time sensor data and system calls.
- * Designed secure transaction workflows and API integrations for distributed systems, Collaborated in design discussions around system architecture, state management, and real-time data handling.

– **TCR Sequence Extraction from RNA-seq Data**

Bioinformatics Data Pipeline | Python | R | RStudio

2025

- * Achieved 76% alignment/extraction efficiency in validation runs; implemented cross-tool comparison and overlap visualization to quantify concordance between extraction tools (MiXCR vs TRUST4).
- * Built reproducible notebooks and automated data-download scripts (ENA), caching layers to avoid redundant computation, and end-to-end reporting for downstream analysis and clinical interpretation.